

MID-COURSE ASSIGNMENT • SALESFORCE PLATFORM

Canteen Automation System

Digitizing everyday canteen operations — ordering, payments, approvals, and delivery tracking, all in one connected system.

Project Overview

Objective

Replacing a manual canteen process with a connected, self-service Salesforce system.



Browse & Order Food

Customers browse items by category and add them to a cart.



Online Payment

Orders are paid for online — card or other digital modes.



Order Tracking

Customers get email alerts as their order status changes.



Admin Control

Admin adds, edits, deletes, and enables/disables food items.



Delivery & Visibility

Admin updates delivery status; payments & orders stay fully visible.

Data Model

Six objects capture people, orders, payment, and delivery end-to-end.

■ Admin

`Admin__c`

Manages food items — insert, edit, delete.

■ Customer

`Customer__c`

Auto-numbered ID; name, mobile, email, department.

■ Order Details

`Order_Details__c`

Items ordered; linked to Customer; 15-min prep.

■ Order Summary

`Order_Summary__c`

Status (Available/Not Available) & total amount.

■ Delivery

`Delivery__c`

Ready / Delivered / Not Delivered, by department.

■ Payment

`Payment__c`

Linked to Customer; online payment mode only.

Relationships: Order Details → Customer • Delivery → Order Details • Payment → Customer

What You'll Build

Seven Salesforce building blocks turn the requirements into a working system.

1

Record Types & Page Layouts

Veg / Non-Veg sections with tailored layouts.

2

Registration Email Automation

Confirm registration via Workflow / Process Builder.

3

Object Relationships

Connect Customer, Order, Delivery & Payment.

4

Approval Process

Order Details routed to Admin for finalization.

5

Order Status Email Alert

Notify customer once the order is finalized.

6

Validation Rules & Formula Fields

Enforce data quality and derive key values.

7

Delivery Email Alert

Notify customer on successful delivery.

Record Types & Page Layouts

Separate Veg and Non-Veg sections so each shows only the fields it needs.

1 Two Record Types

"Veg" and "Non-Veg" record types created on the Food Item object.

2 Matching Page Layouts

Each record type is assigned its own page layout.

3 Clean, Focused UI

Irrelevant fields are hidden so staff see only what applies.

4 Usability First

Layout order matches how admin actually enters food items.

Record Types & Page Layouts – Order Details

1

Record Types (Order Details)

SetupHomeObject Manager

SETUP > OBJECT MANAGER

Order Details

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

Record Type

Regular Order

Back to Custom Object: Order Details

Record Type LabelRegular Order

Record Type NameRegular_Order

Namespace Prefix

Description

Active

Created ByJay Shankar Das, 10/2/2026, 2:07 AM

Modified ByJay Shankar Das, 10/2/2026, 2:07 AM

Picklists Available for Editing

ActionFieldModified Date

EditOrder Status10/2/2026, 2:07 AM

2

Page Layouts (Order Details)

SetupHomeObject Manager

SETUP > OBJECT MANAGER

Order Details

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

SaveQuick SavePreview As...CancelLayout Properties

Fields

Buttons

Quick Actions

Mobile & Lightning Actions

Expanded Lookups

Related Lists

Quick FindField Name

Section Space

Order Details Name

Quantity

Estimated Delivery...

Order Items

Record Type

Last Modified By

Order Status

Order Date

Owner

Created By

Admin

Order Details Detail

Standard Buttons

EditDeleteCloneChange OwnerChange Record Type

Printable ViewSharingSharing HierarchyEdit LabelsReassign

Information

Order Details NameGEN-2004-001234CustomerSample Text

Order ItemsSample TextRecord TypeRegular Order

Quantity363OwnerSample Text

Order StatusSample TextAdminSample Text

Order Date10/2/2026, 2:10 AM

Estimated Delivery Time10/2/2026, 2:10 AM

System Information

Created BySample Text

Last Modified BySample Text

Canteen Automation System

5

Registration Email Automation

Workflow Rule or Process Builder sends a confirmation the moment a customer registers.

1 Trigger

Fires when a new Customer record is created.

2 Condition

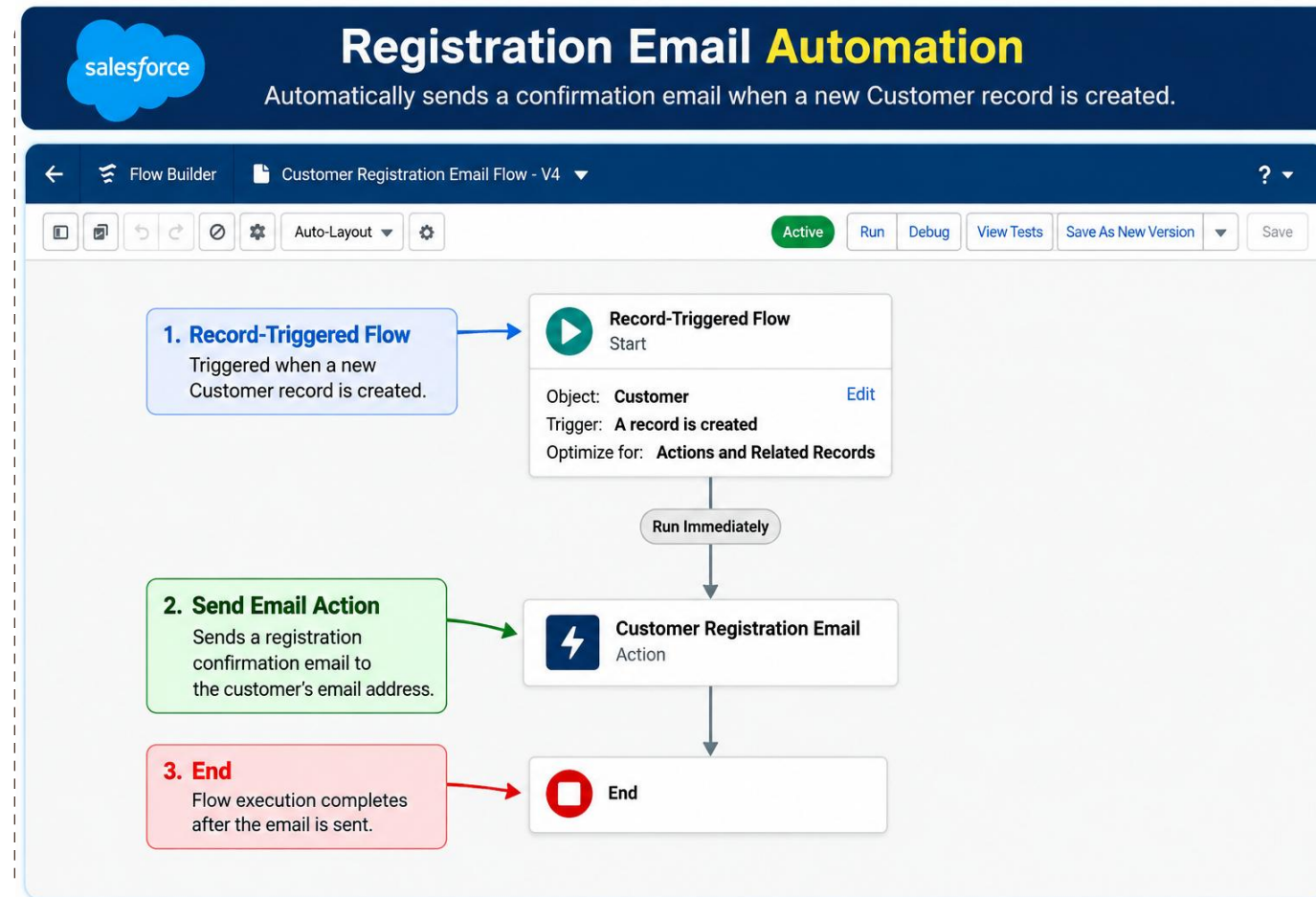
Runs once — on initial registration only.

3 Action

Sends a confirmation email alert to the customer's email address.

4 Benefit

Customer knows immediately that their registration succeeded.



Object Relationships

Lookup relationships tie every stage of the order lifecycle together.

1 **Order Details → Customer**

Every order traces back to the customer who placed it.

2 **Delivery → Order Details**

Delivery status is tracked against a specific order.

3 **Payment → Customer**

Payments are linked to the paying customer.

4 **Why It Matters**

One connected record lets admin see a customer's full journey.

salesforce

Object Relationships

Shows how different objects are connected in the Canteen Automation System.

SetupHomeObject Manager

SETUP > OBJECT MANAGER

Order Details

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

7 Items, Sorted by Field Label

Quick Find

NewDeleted FieldsField Dependencies

Field Label	Field Name	Data Type	Relationship
Customer	Customer__c	Lookup(Customer)	Many Orders → One Customer
Food Item	Food_Item__c	Lookup(Food Item)	Many Orders → One Food Item
Payment	Payment__c	Lookup(Payment)	Many Orders → One Payment
Delivery	Delivery__c	Lookup(Delivery)	Many Orders → One Delivery
Owner	OwnerId	Lookup(User)	Many Orders → One User
Created By	CreatedById	Lookup(User)	Many Orders → One User
Last Modified By	LastModifiedById	Lookup(User)	Many Orders → One User

Relationship Diagram

Customer

Customer Name

Email

Food Item

Food Item Name

Price

Order Details

Customer__c

Food_Item__c

Payment__c

Delivery__c

Payment

Payment Date

Amount

Delivery

Delivery Date

Delivery Status

Canteen Automation System

7

Approval Process — Order Finalization

Every order is routed to the Admin for approval before it's finalized.

1 Submission

An Order Details record is submitted for approval by the customer's order.

2 Approver

Routed to the Admin (Canteen) for review.

3 Decision

Admin approves or rejects based on item availability.

4 On Approval

Order Summary is finalized and moves to Delivery.

salesforce

Approval Process — Order Finalization

Ensures that an order is approved before it is finalized for processing.

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Order Details

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

Triggers

Approval Processes

Flow Triggers

Approval Process Detail

EditCloneDeactivateViewDiagram

Process Name	Order Finalization
Unique Name	Order_Finalization
Description	Approves the order before finalization for processing.
Object	Order Details
Active	✓
Created By	Jay Shankar Das, 9/10/2026, 11:25 AM
Modified By	Jay Shankar Das, 9/10/2026, 11:40 AM

Approval Steps

Edit

Action	Step Number	Name	Approver	Criteria
Edit Del	1	Manager Approval	Order Manager (Queue/User)	Order Amount > 500
Edit Del	2	Admin Approval	System Administrator	Order Amount > 2000

Final Approval Actions

Edit

Action Type	Update a Record
Field to Update	Order Status
New Field Value	Approved

Approval Process Flow — Order Finalization

Order Created

Order Status = Pending Approval

Step 1

Manager Approval

Criteria: Order Amount > 500

Step 2

Admin Approval

Criteria: Order Amount > 2000

Final Action

Update Record

Order Status = Approved

Order Finalized

Ready for Processing

Approval Criteria

- Order Amount > 500 → Manager Approval
- Order Amount > 2000 → Admin Approval

Final Approval Action

- Update Order Status to Approved
- Order is finalized for further processing

Salesforce Concepts Used

- Approval Process
- Approval Steps
- Field Update Action
- Record Status Management

Canteen Automation System

8

Order Status & Payment Email Alert

Customers are emailed as soon as their order is finalized.

1 Trigger Point

Fires once the Approval Process finalizes the order.

2 Content

Includes order status and payment details.

3 Channel

Sent as an Email Alert tied to the Order Summary record.

4 Outcome

Customer always knows exactly where their order stands.

Order Status & Payment Email Alert

Automatically sends email notifications when the Order Status or Payment Status is updated.

1 Record-Triggered Flow

Record-Triggered Flow Start

Object: Order Details

Trigger: A record is updated

Condition: Order Status OR Payment Status Is Changed

Send Email Alert Action

End

2 Flow Trigger Condition

Entry Conditions

Condition Requirements

Any Condition Is Met (OR)

1

Order_Status__c

Is Changed

True

2

Payment_Status__c

Is Changed

True

Add Condition

3 Send Email Alert Action

Send Email

Label: Order Status & Payment Email Alert

API Name: Order_Status_Payment_Email_Alert

Email Template: Order Notification Email

Recipient Type: User or Contact

Recipient Id: {!Order_Details__c.Customer__c.Email}

Subject: Order Update: {!Order_Details__c.Order_Status__c}

Body: Dear Customer,
Your order status has been updated to: {!Order_Details__c.Order_Status__c}
Payment Status: {!Order_Details__c.Payment_Status__c}
Thank you for using our Canteen Service.
Regards,
Canteen Team

4 Sample Email Output

Order Update: Approved

Canteen Service

Dear Customer,

Your order status has been updated to: **Approved**

Payment Status: **Paid**

Thank you for using our Canteen Service.

Regards,
Canteen Team

Validation Rules & Formula Fields

Keeps data accurate and automatically derives values admins shouldn't type by hand.

1 Validation Rule

Blocks payment records unless Mode of Payment is Online.

2 Validation Rule

Prevents an order from saving without at least one item selected.

3 Formula Field

Total Amount calculated automatically from ordered items.

4 Formula Field

Estimated ready-time derived as Order Time + 15 minutes.

salesforce

Validation Rules & Formula Fields

Ensure data accuracy and calculate values automatically in the Canteen Automation System.

1 Validation Rule – Order Quantity

SetupHomeObject Manager

SETUP > OBJECT MANAGER > Order Details

Order Details

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Validation Rules

Triggers

Flow Triggers

Object Access

Order Quantity Validation

Rule Name

Order_Quantity_Validation

Error Condition Formula

AND(
NOT(ISBLANK(Quantity__c)),
Quantity__c <= 0
)

Error Message

Quantity must be greater than 0.

Error Location

Quantity

Description

Ensures that the order quantity entered is a positive value.

Active

✓

Created By

Jay Shankar Das, 9/12/2026, 10:15 AM

Modified By

Jay Shankar Das, 9/12/2026, 10:20 AM

What it does?

This validation rule prevents users from saving a record if the Order Quantity is 0 or less.

2 Formula Field – Total Amount

SetupHomeObject Manager

SETUP > OBJECT MANAGER > Order Details

Order Details

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Validation Rules

Triggers

Flow Triggers

Object Access

Total Amount

Field Label

Total Amount

Field Name

Total_Amount__c

Data Type

Formula (Currency)

Formula

Quantity__c * Food_Item__r.Price__c

Description

Calculates the total amount for the order based on quantity and food item price.

Active

✓

Created By

Jay Shankar Das, 9/12/2026, 10:25 AM

Modified By

Jay Shankar Das, 9/12/2026, 10:25 AM

What it does?

This formula field automatically calculates the total amount by multiplying Quantity with Food Item Price.

Canteen Automation System

10

Delivery Confirmation Email Alert

A final email confirms the order reached the customer's department.

1 Trigger Point

Fires when Delivery Status changes to "Delivered".

2 Destination

Sent to the customer's registered email address.

3 Content

Confirms successful delivery to their department address.

4 Closes the Loop

Customer has a clear record from order to doorstep.

salesforce

Delivery Confirmation Email Alert

Automatically sends an email to the customer when the order is delivered.

1

Record-Triggered Flow

←

Flow Builder

Delivery Confirmation Email Flow - V1

↶

🔍

🔗

📄

🔄

Auto-Layout ▾

⚙️

Active

Run

Debug

Save As New Version

▶️

Record-Triggered Flow

Start

Object: Order Details

Trigger: A record is updated

Condition: Order Status Equals Delivered

⊕

✉️

Send Delivery Confirmation Email

Action

⊕

⏹️

End

2

Flow Trigger Condition

Entry Conditions

Condition Requirements

All Conditions Are Met (AND) ▾

1

Field

Order_Status__c

Operator

Equals ▾

Value

Delivered

🗑️

2

Field

Delivery_Status__c

Operator

Equals ▾

Value

Delivered

🗑️

+ Add Condition

3

Send Email Alert Action

✉️

Send Email

🗑️

Label

Delivery Confirmation Email

API Name

Delivery_Confirmation_Email

Email Template

Order Delivered Email 🔍

Recipient Type

User or Contact ▾

Recipient Id

{!Order_Details__c.Customer__c.Email}

Subject

Your Canteen Order Has Been Delivered

Body

Dear Customer,
Your order has been successfully delivered.
Order Number: {!Order_Details__c.Order_Details_Name__c}
Delivered On: {!Order_Details__c.Delivery_Date__c}

Thank you for using our Canteen Service.
Regards,
Canteen Team

4

Sample Delivery Confirmation Email

Your Canteen Order Has Been Delivered

Inbox x

🖨️

✉️

Canteen Service

to me ▾

10:24 AM (0 minutes ago) ☆ ↶ ⋮

Dear Customer,

Your order has been successfully delivered.

Order Number: **ORD-2026-001234**

Delivered On: **Sep 13, 2026**

Thank you for using our Canteen Service.

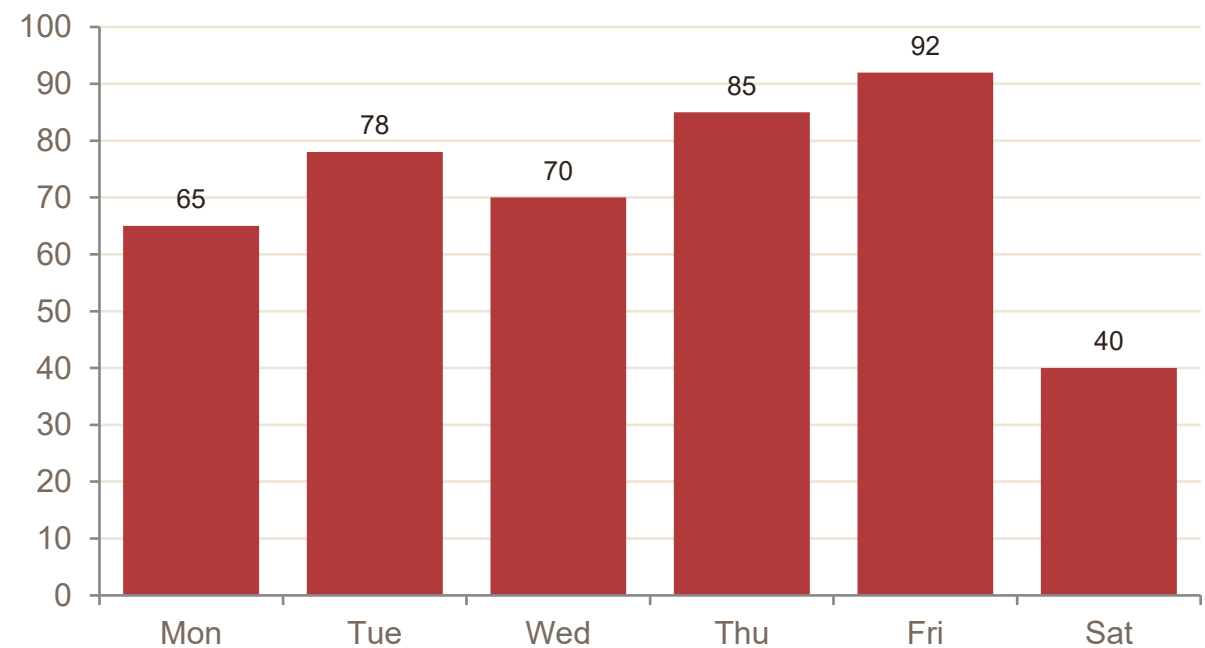
Regards,

Canteen Team

Reports & Dashboards

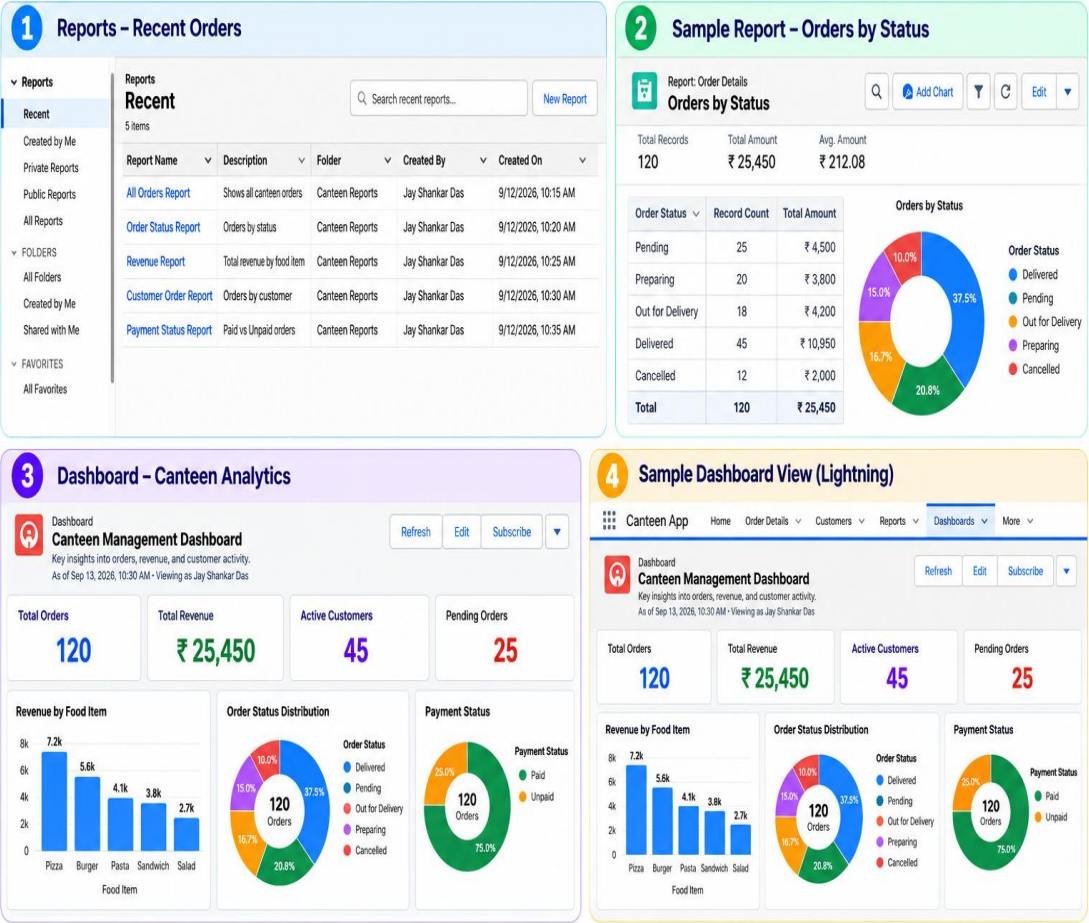
Operational visibility into daily orders, delivery performance, and food demand.

Orders per Day (Illustrative)



Reports & Dashboards

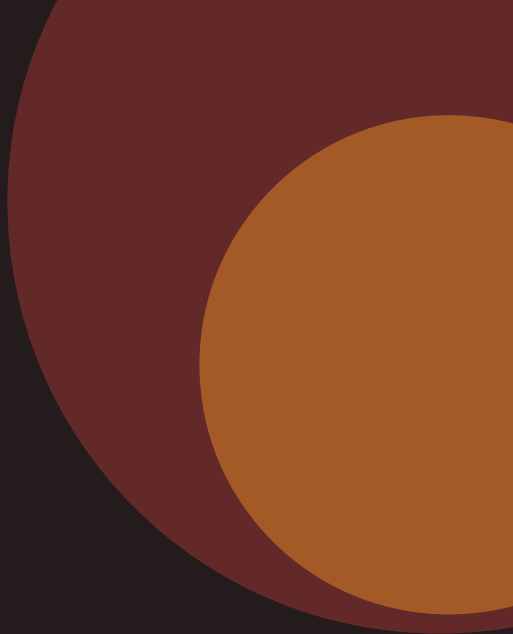
Provides meaningful insights into orders, revenue, and customer activity in the Canteen Automation System.



Why This Matters

- Manual canteen registers are replaced by one connected system.
- Approval routing keeps every order accountable before it's finalized.
- Automated emails keep customers informed at every stage.
- Validation rules and formulas keep data clean without manual checks.

Ready for implementation and review.



Thank You

Canteen Automation System

Questions & Discussion

